

SQL Queries (Midterm)

Single-Row Functions

1. Display ENAME in lowercase, JOB with the first letter capitalized, and HIREDATE formatted as 'Month DD, YYYY' from the "emp" table.
 2. Calculate the total income (SAL + COMM) for each employee, replacing NULL COMM values with 0, from the "emp" table.
 3. Show ENAME and the number of months since HIREDATE as of April 10, 2025, from the "emp" table.
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Multiple-Row (Aggregate) Functions

4. Find the minimum SAL, maximum SAL, and average SAL across all employees in the "emp" table.
 5. Calculate the total COMM paid to employees in DEPTNO 30 from the "emp" table.
 6. Count the number of employees with a non-NULL COMM in the "emp" table.
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GROUP BY

7. Display the average SAL for each JOB in the "emp" table.
 8. Show the total SAL and number of employees for each DEPTNO in the "emp" table.
 9. Find the maximum SAL for each GENDER within each DEPTNO in the "emp" table.
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HAVING Clause

10. List JOB titles from the "emp" table with more than 3 employees and an average SAL greater than 2000.
 11. Show DEPTNO values from the "emp" table where the total SAL exceeds 10000.
 12. Find JOB titles from the "emp" table with at least 2 employees and a minimum SAL above 1000.
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Wildcards

13. Retrieve ENAME and JOB for employees whose ENAME contains 'A' anywhere in the "emp" table.
 14. List ENAME for employees whose JOB ends with 'MAN' from the "emp" table.
 15. Display ENAME and DEPTNO for employees whose ENAME starts with 'S' and has exactly 6 letters in the "emp" table.
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ORDER BY

16. Show ENAME, SAL, and DEPTNO from the "emp" table, sorted by SAL in ascending order and then ENAME in descending order.
 17. List JOB and HIREDATE from the "emp" table, ordered by HIREDATE from oldest to newest.
 18. Display ENAME and COMM from the "emp" table, sorted by COMM in descending order, with NULLs appearing last.
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JOINS

19. Write a query to display ENAME, JOB, and DNAME for all employees using an INNER JOIN between "emp" and "dept".
 20. List all ENAME and DNAME values, including employees without a department, using a LEFT JOIN between "emp" and "dept".
 21. Show all DNAME and ENAME values, including departments without employees, using a RIGHT JOIN between "emp" and "dept".
 22. Display ENAME and DNAME for all possible combinations, including unmatched rows, using a FULL JOIN between "emp" and "dept".
 23. Generate all possible pairs of ENAME and DNAME using a CROSS JOIN between "emp" and "dept".
 24. Show each employee's ENAME and their manager's ENAME using a SELF JOIN on the "emp" table.
 25. Find ENAME and DNAME for employees in departments located in 'CHICAGO' using an INNER JOIN between "emp" and "dept".
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Set Operators

26. Combine ENAME and DEPTNO for employees in DEPTNO 20 with those in DEPTNO 30, removing duplicates, using UNION between two queries on the "emp" table.
 27. List ENAME for employees in DEPTNO 10 and DEPTNO 20, showing only names common to both, using INTERSECT between two queries on the "emp" table.
 28. Show ENAME for employees in DEPTNO 30 that are not in DEPTNO 20 using MINUS (or EXCEPT) between two queries on the "emp" table.
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Subqueries

29. Display ENAME and SAL for employees earning more than the average SAL of all employees in the "emp" table.
 30. List ENAME and JOB for employees in the same DEPTNO as 'HAIDER' using a subquery on the "emp" table.
 31. Find ENAME and DEPTNO for employees whose SAL is greater than the maximum SAL in DEPTNO 10 using a subquery on the "emp" table.
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Mixed Concepts

32. Show the top 5 ENAME values in uppercase, their DNAME, and total income (SAL + COMM, with COMM NULLs as 0), sorted by total income descending, using an INNER JOIN between "emp" and "dept".
33. Find DEPTNO and average SAL for departments with more than 2 employees whose DNAME starts with 'S', using an INNER JOIN between "emp" and "dept".
34. List ENAME and HIREDATE (formatted as 'DD-MMM-YY') for employees hired after the earliest HIREDATE in DEPTNO 20, sorted by HIREDATE, from the "emp" table.